Untitled1

December 4, 2025

[1]:
```
#  1
import cvxpy as cp
x_1, x_2 = cp.Variable(), cp.Variable()
obj = cp.Maximize(20*x_1 + 60*x_2)
cons = [5*x_1 + 4*x_2 <= 80, 2*x_1 + 4*x_2 <= 40, 2*x_1 + 8*x_2 <= 64, x_1 >=␣
 ↪0, x_2 >= 0]
P = cp.Problem(obj, cons)
P.solve()
print(P.value, x_1.value, x_2.value)
```

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519.9999999974734 7.999999999864201 6.000000000003156
```

[7]:
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#  2
def objective_function(x_1, x_2):
    return 2*x_1**2 - 2*x_1*x_2 + (1 / 3)*x_2**3 - x_2**2

def gradient(x_1, x_2):
    return 4*x_1 - 2*x_2, -2*x_1 + x_2**2 - 2*x_2

x_1, x_2 = 0, 1
a = 0.1

for i in range(51):
    f = objective_function(x_1, x_2)
    print(f"Iteration:{i}, x_1 = {x_1}, x_2 = {x_2}, f = {f}")

    grad_x1, grad_x2 = gradient(x_1, x_2)

    x_1 = x_1 - a * grad_x1
    x_2 = x_2 - a * grad_x2
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Iteration:0, x_1 = 0, x_2 = 1, f = -0.6666666666666667
Iteration:1, x_1 = 0.2, x_2 = 1.1, f = -1.1263333333333334
Iteration:2, x_1 = 0.34, x_2 = 1.239, f = -1.5124360270000001
Iteration:3, x_1 = 0.45180000000000003, x_2 = 1.4012879000000003, f =
-1.9043717715866957
Iteration:4, x_1 = 0.5513375800000001, x_2 = 1.5755447021313593, f =
-2.3080290666019074
Iteration:5, x_1 = 0.645911488426272, x_2 = 1.7526870477162118, f =
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-2.7069699261159768
Iteration:6, x_1 = 0.7380843025990056, x_2 = 1.9252155662214916, f =
-3.080285939328144
Iteration:7, x_1 = 0.8278936948037017, x_2 = 2.087230042343437, f =
-3.4106962497780855
Iteration:8, x_1 = 0.9141822253509084, x_2 = 2.2346018648067663, f =
-3.688199618605638
Iteration:9, x_1 = 0.9954297081718984, x_2 = 2.3650141334185135, f =
-3.9105371405538376
Iteration:10, x_1 = 1.0702606515868418, x_2 = 2.4777737166096636, f =
-4.08152386955605
Iteration:11, x_1 = 1.1377111342740378, x_2 = 2.5734443311767983, f =
-4.208537381333098
Iteration:12, x_1 = 1.1973155467997822, x_2 = 2.653413851700366, f =
-4.300210340370567
Iteration:13, x_1 = 1.2490720984199426, x_2 = 2.7194992245608582, f =
-4.364836612705298
Iteration:14, x_1 = 1.2933431039641372, x_2 = 2.7736458859183077, f =
-4.409534602379015
Iteration:15, x_1 = 1.3307350395621438, x_2 = 2.817732533847641, f =
-4.439977274037332
Iteration:16, x_1 = 1.3619875305068145, x_2 = 2.853464385299253, f =
-4.460456065650218
Iteration:17, x_1 = 1.3878853953639394, x_2 = 2.8823288686433424, f =
-4.474096056200076
Iteration:18, x_1 = 1.4091970109470322, x_2 = 2.9055897507433177, f =
-4.483109011827562
Iteration:19, x_1 = 1.4266361567168828, x_2 = 2.924301923118926, f =
-4.489026687352946
Iteration:20, x_1 = 1.440842078653915, x_2 = 2.939335365330383, f =
-4.492892291544404
Iteration:21, x_1 = 1.4523723202584256, x_2 = 2.951401615139053, f =
-4.495407118780148
Iteration:22, x_1 = 1.461703715182866, x_2 = 2.9610792528340077, f =
-4.497037832757254
Iteration:23, x_1 = 1.4692380796765212, x_2 = 2.968836812280982, f =
-4.498092486957397
Iteration:24, x_1 = 1.4753102102621092, x_2 = 2.9750525888770123, f =
-4.498773150126557
Iteration:25, x_1 = 1.480196643932668, x_2 = 2.9800313580464555, f =
-4.499211707250682
Iteration:26, x_1 = 1.4841242579688918, x_2 = 2.98401826894826, f =
-4.499493894515701
Iteration:27, x_1 = 1.487278208570987, x_2 = 2.9872102713899933, f =
-4.499675271618612
Iteration:28, x_1 = 1.4898089794205909, x_2 = 2.9897654468324015, f =
-4.499791752469027
Iteration:29, x_1 = 1.4918384770188349, x_2 = 2.991810589375705, f =
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-4.499866505396757
Iteration:30, x_1 = 1.493465204086442, x_2 = 2.993447342384553, f =
-4.499914452586026
Iteration:31, x_1 = 1.4947685909287758, x_2 = 2.9947571525158376, f =
-4.499945192816054
Iteration:32, x_1 = 1.495812585060433, x_2 = 2.9958052609502834, f =
-4.499964894270101
Iteration:33, x_1 = 1.4966486032263164, x_2 = 2.996643913998687, f =
-4.499977517404644
Iteration:34, x_1 = 1.4973179447355274, x_2 = 2.997314942713151, f =
-4.499985603490582
Iteration:35, x_1 = 1.4978537553839466, x_2 = 2.997851833621733, f =
-4.499990782315891
Iteration:36, x_1 = 1.4982826199547146, x_2 = 2.99828138978795, f =
-4.499994098675249
Iteration:37, x_1 = 1.4986258499304188, x_2 = 2.9986250625016067, f =
-4.499996222124261
Iteration:38, x_1 = 1.4989005224585725, x_2 = 2.998900018441735, f =
-4.4999975816330995
Iteration:39, x_1 = 1.4991203171634906, x_2 = 2.9991199945608127, f =
-4.499998451975593
Iteration:40, x_1 = 1.499296189210257, x_2 = 2.9992959827282286, f =
-4.499999009126324
Iteration:41, x_1 = 1.4994369100718, x_2 = 2.9994367779149567, f =
-4.499999365770153
Iteration:42, x_1 = 1.4995495016260714, x_2 = 2.9995494170414223, f =
-4.499999594056698
Iteration:43, x_1 = 1.4996395843839272, x_2 = 2.9996395302475674, f =
-4.499999740177754
Iteration:44, x_1 = 1.4997116566798698, x_2 = 2.9997116220314815, f =
-4.499999833704271
Iteration:45, x_1 = 1.4997693184142182, x_2 = 2.9997692962386777, f =
-4.499999893565873
Iteration:46, x_1 = 1.4998154502962664, x_2 = 2.9998154361036278, f =
-4.49999993187967
Iteration:47, x_1 = 1.4998523573984854, x_2 = 2.9998523483150468, f =
-4.499999956401717
Iteration:48, x_1 = 1.4998818841021007, x_2 = 2.9998818782886234, f =
-4.499999972096446
Iteration:49, x_1 = 1.499905506118985, x_2 = 2.9999055023983203, f =
-4.499999982141392
Iteration:50, x_1 = 1.499924404151055, x_2 = 2.9999244017698095, f =
-4.499999988570321
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